CHAPTER 10

Nitrogen- and Phosphorus-Containing Compounds

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

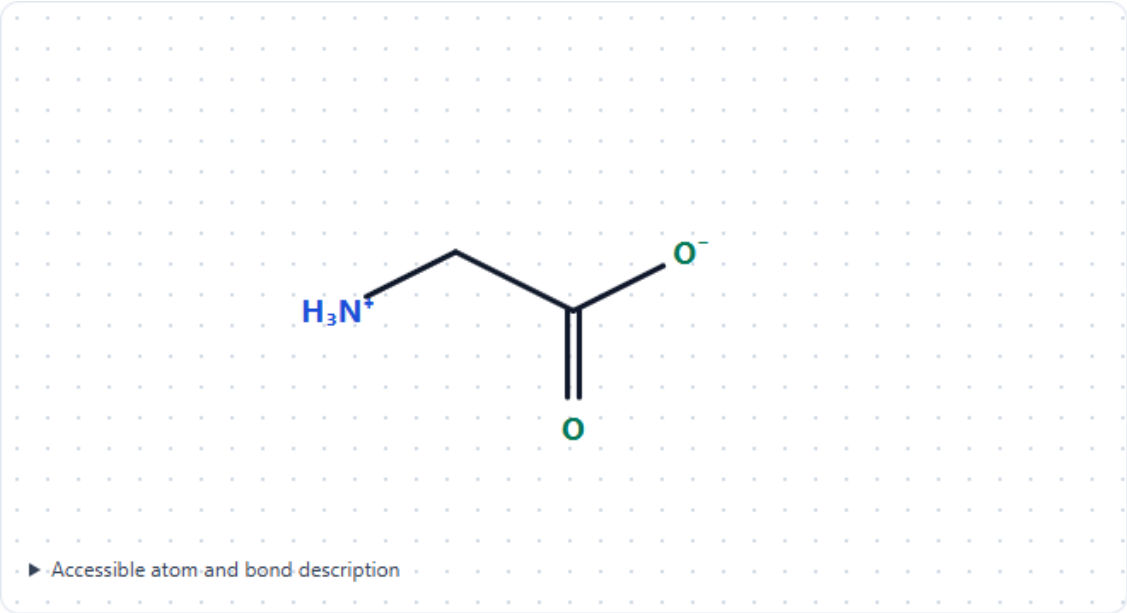
Source snapshot SHA-256: c79826a794164a21394863004b938faf59738f32cccc97169d32a44eb89edb60

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

glycine-charge | Independent exercise

Screenshot: glycine-charge-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=45.00, y=115.00; charge=1; explicit H override=none

1 (B): C; x=105.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=165.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=165.00, y=175.00; charge=0; explicit H override=none

4 (E): O; x=225.00, y=85.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1]]

glycine-charge:stage-1 | What is the net formal charge of the displayed zwitterion?

Answer type: choice

Option 0 (A): 0

Option 1 (B): +1

Option 2 (C): −1

glycine-charge:stage-2 | Why is glycine achiral?

Answer type: choice

Option 0 (A): Its alpha carbon has two H substituents

Option 1 (B): All amino acids are achiral

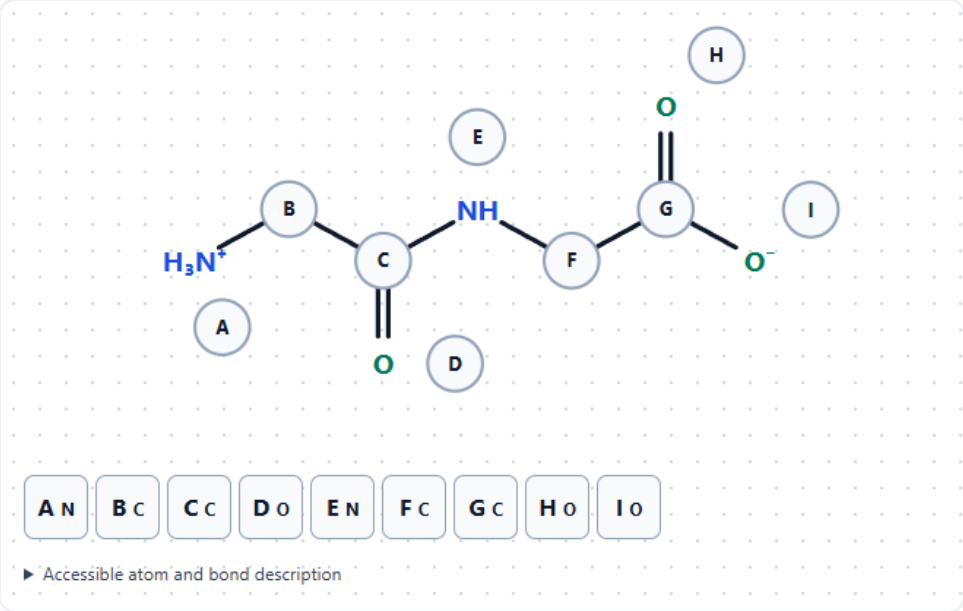
Option 2 (C): Its nitrogen is neutral

glycine-charge:stage-3 | Using model pKa values 2.30 and 9.60, calculate pI. Report two decimal places.

Answer type: number | unit: pH units

peptide-link | Independent exercise

Screenshot: peptide-link-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=35.00, y=110.00; charge=1; explicit H override=none

1 (B): C; x=90.00, y=80.00; charge=0; explicit H override=none

2 (C): C; x=145.00, y=110.00; charge=0; explicit H override=none

3 (D): O; x=145.00, y=170.00; charge=0; explicit H override=none

4 (E): N; x=200.00, y=80.00; charge=0; explicit H override=none

5 (F): C; x=255.00, y=110.00; charge=0; explicit H override=none

6 (G): C; x=310.00, y=80.00; charge=0; explicit H override=none

7 (H): O; x=310.00, y=20.00; charge=0; explicit H override=none

8 (I): O; x=365.00, y=110.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 2], [6, 8, 1]]

peptide-link:stage-1 | Select the two endpoints of the internal peptide C–N bond.

Answer type: atoms

peptide-link:stage-2 | Why is rotation around this bond restricted?

Answer type: choice

Option 0 (A): Amide resonance gives partial double-bond character

Option 1 (B): Every single bond is completely rigid

Option 2 (C): The bond is an ionic salt bridge

peptide-link:stage-3 | Peptide sequences are read from…

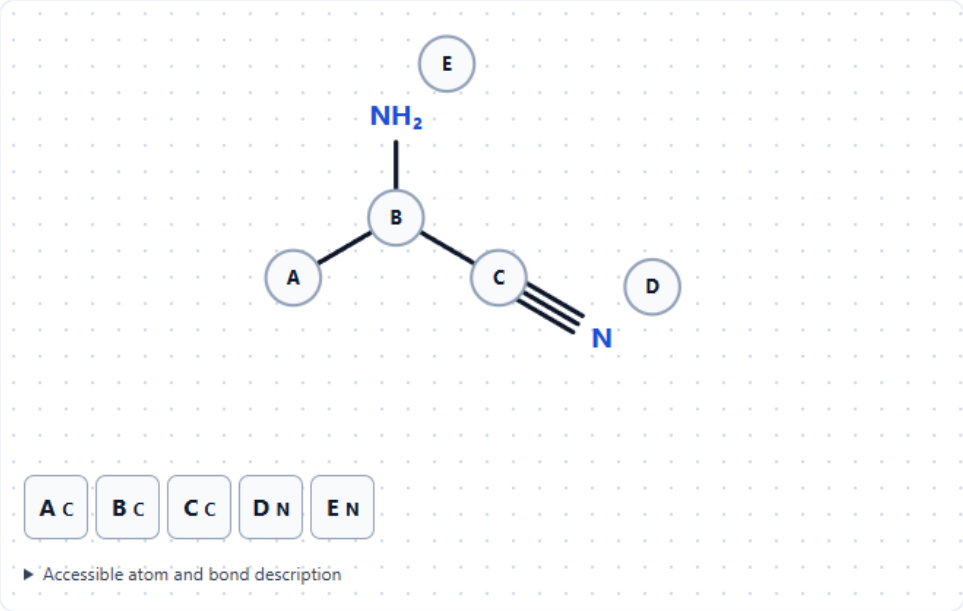
Answer type: choice

Option 0 (A): N-terminus to C-terminus

Option 1 (B): C-terminus to N-terminus by default

strecker-map | Independent exercise

Screenshot: strecker-map-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=110.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=170.00, y=120.00; charge=0; explicit H override=none

3 (D): N; x=230.00, y=155.00; charge=0; explicit H override=none

4 (E): N; x=110.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 3], [1, 4, 1]]

strecker-map:stage-1 | Select the carbon that becomes the carboxyl carbon after hydrolysis.

Answer type: atoms

strecker-map:stage-2 | Which group is retained as the alanine side chain?

Answer type: choice

Option 0 (A): Methyl

Option 1 (B): Nitrile nitrogen

Option 2 (C): A new phenyl group

strecker-map:stage-3 | Under achiral conditions, the chiral product is generally…

Answer type: choice

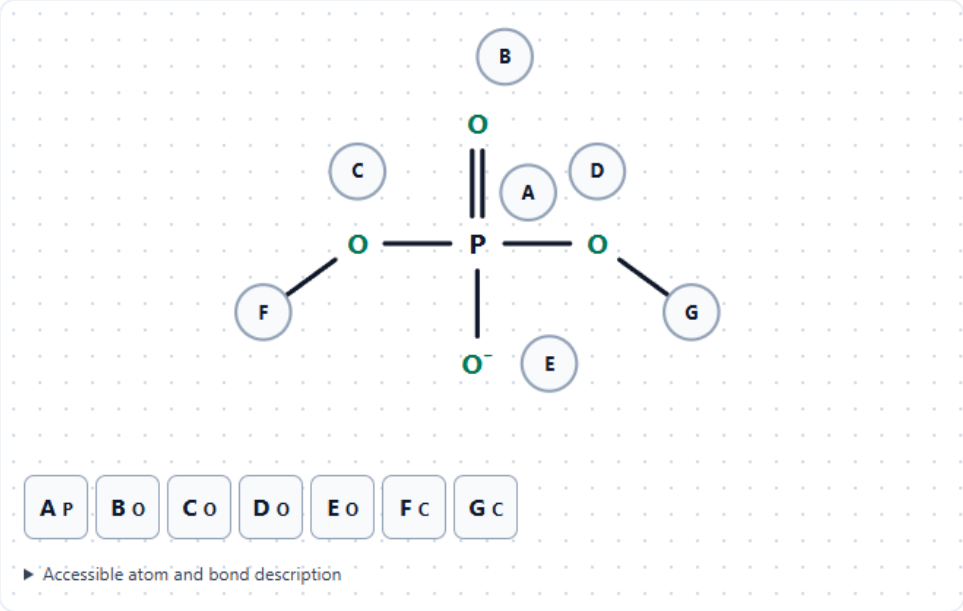
Option 0 (A): Racemic

Option 1 (B): Only L without a chiral influence

Option 2 (C): Only D without a chiral influence

phosphate-links | Independent exercise

Screenshot: phosphate-links-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): P; x=160.00, y=100.00; charge=0; explicit H override=none

1 (B): O; x=160.00, y=30.00; charge=0; explicit H override=none

2 (C): O; x=90.00, y=100.00; charge=0; explicit H override=none

3 (D): O; x=230.00, y=100.00; charge=0; explicit H override=none

4 (E): O; x=160.00, y=170.00; charge=-1; explicit H override=none

5 (F): C; x=35.00, y=140.00; charge=0; explicit H override=none

6 (G): C; x=285.00, y=140.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [0, 2, 1], [0, 3, 1], [0, 4, 1], [2, 5, 1], [3, 6, 1]]

phosphate-links:stage-1 | Select the phosphorus atom.

Answer type: atoms

phosphate-links:stage-2 | This structure models a…

Answer type: choice

Option 0 (A): Phosphodiester

Option 1 (B): Phosphoanhydride

Option 2 (C): Carboxylic anhydride

phosphate-links:stage-3 | Why can ATP hydrolysis be favorable overall?

Answer type: choice

Option 0 (A): Products have lower free energy under the stated conditions

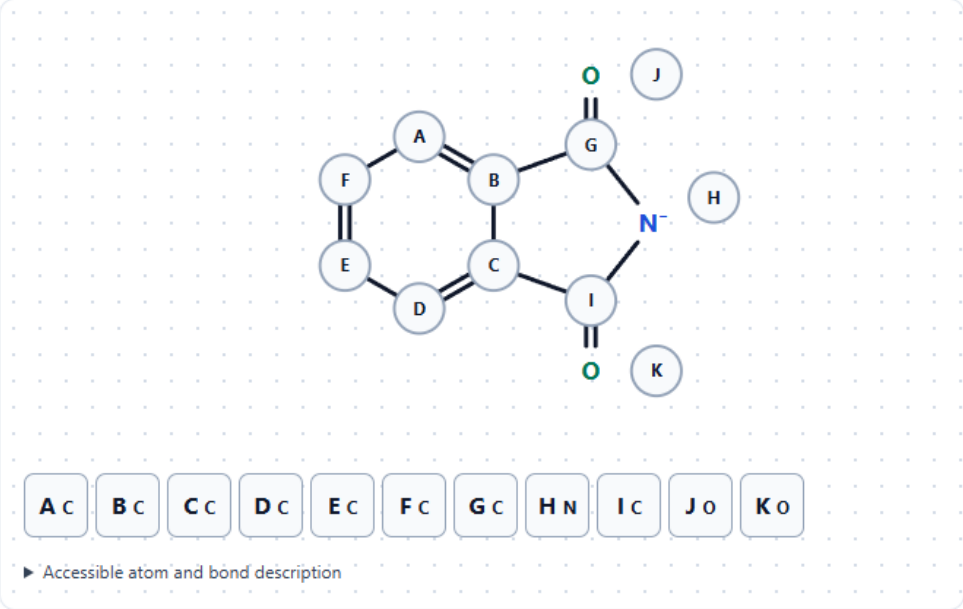
Option 1 (B): Breaking a bond itself releases energy

Option 2 (C): All phosphate bonds break spontaneously at the same rate

gabriel-route | Independent exercise

The canvas shows the protected nitrogen reagent, with its potassium counterion omitted; later questions refer to the route described in Learn & Explore.

Screenshot: gabriel-route-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=130.00, y=45.00; charge=0; explicit H override=none

1 (B): C; x=177.63, y=72.50; charge=0; explicit H override=none

2 (C): C; x=177.63, y=127.50; charge=0; explicit H override=none

3 (D): C; x=130.00, y=155.00; charge=0; explicit H override=none

4 (E): C; x=82.37, y=127.50; charge=0; explicit H override=none

5 (F): C; x=82.37, y=72.50; charge=0; explicit H override=none

6 (G): C; x=240.00, y=50.00; charge=0; explicit H override=none

7 (H): N; x=280.00, y=100.00; charge=-1; explicit H override=0

8 (I): C; x=240.00, y=150.00; charge=0; explicit H override=none

9 (J): O; x=240.00, y=5.00; charge=0; explicit H override=none

10 (K): O; x=240.00, y=195.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [1, 6, 1], [6, 7, 1], [7, 8, 1], [8, 2, 1], [6, 9, 2], [8, 10, 2]]

gabriel-route:stage-1 | Select the nitrogen nucleophile of the displayed phthalimide anion.

Answer type: atoms

gabriel-route:stage-2 | After protected aminomalonate formation and deprotonation, a suitable primary alkyl halide installs the side chain mainly by…

Answer type: choice

Option 0 (A): SN2 carbon alkylation

Option 1 (B): Electrophilic aromatic substitution on the product amino acid

Option 2 (C): Direct reduction of the ester to an alkane

gabriel-route:stage-3 | After hydrolysis of the malonate esters, how many carboxyl carbons are lost as CO2 in the usual single decarboxylation?

Answer type: number | unit: carboxyl carbons